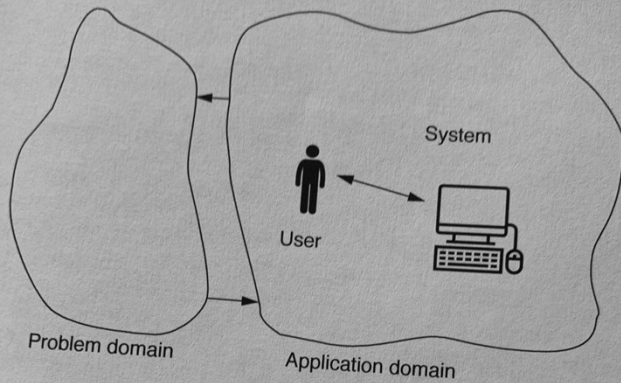




Figur 1.1: The system's context.

n domain is part of the technicians' job, and the problem domain
ludes signals, protocols, lines, transmitters, and
A key tech...

System Development

Session 1: Introduction to SD

Anders Bruun

hcc



Anders

2016-	Associate Professor, AAU
2013 - 2016	Assistant Professor, AAU
2009 - 2012	Ph.D. student, AAU
2007 – 2009	UX Consultant, Mjølner Informatics
2002 – 2007	Informatics student, AAU

Courses	Advanced topics in HCI Basic programming Design and Evaluation of User Interfaces System Development Specialization in HCI User Centered Design
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Course Structure

11 Sessions

Object Oriented
Analysis and Design.
Video material is
available this semester.

Interactive

No lengthy talks, few
slides, emphasis on
exercises

Exam

Written, 7-grade scale,
multiple choice, 3 hrs,
analog/printed aids
only

Goals

This module has the purpose of giving the student knowledge and skills in **modelling** the **structure** and **behaviour** of objects within problem and application domains. Models will be applied to create a **component design** specifying the **architecture** of a given system.

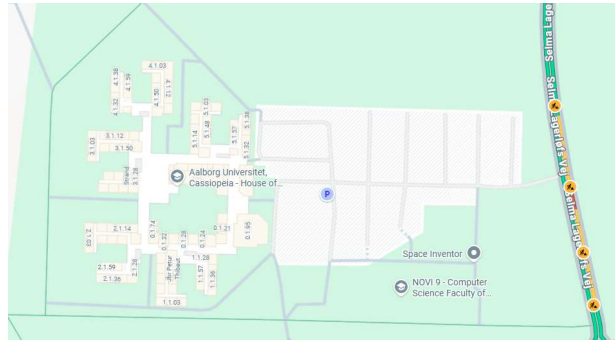
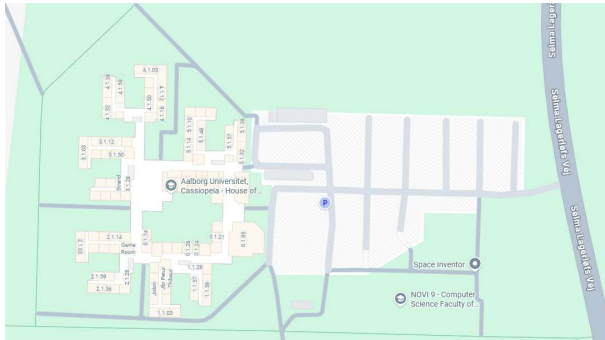
What's a Model?

A representation of a part of the real world

Emphasises certain aspects, i.e. those useful for the current purpose

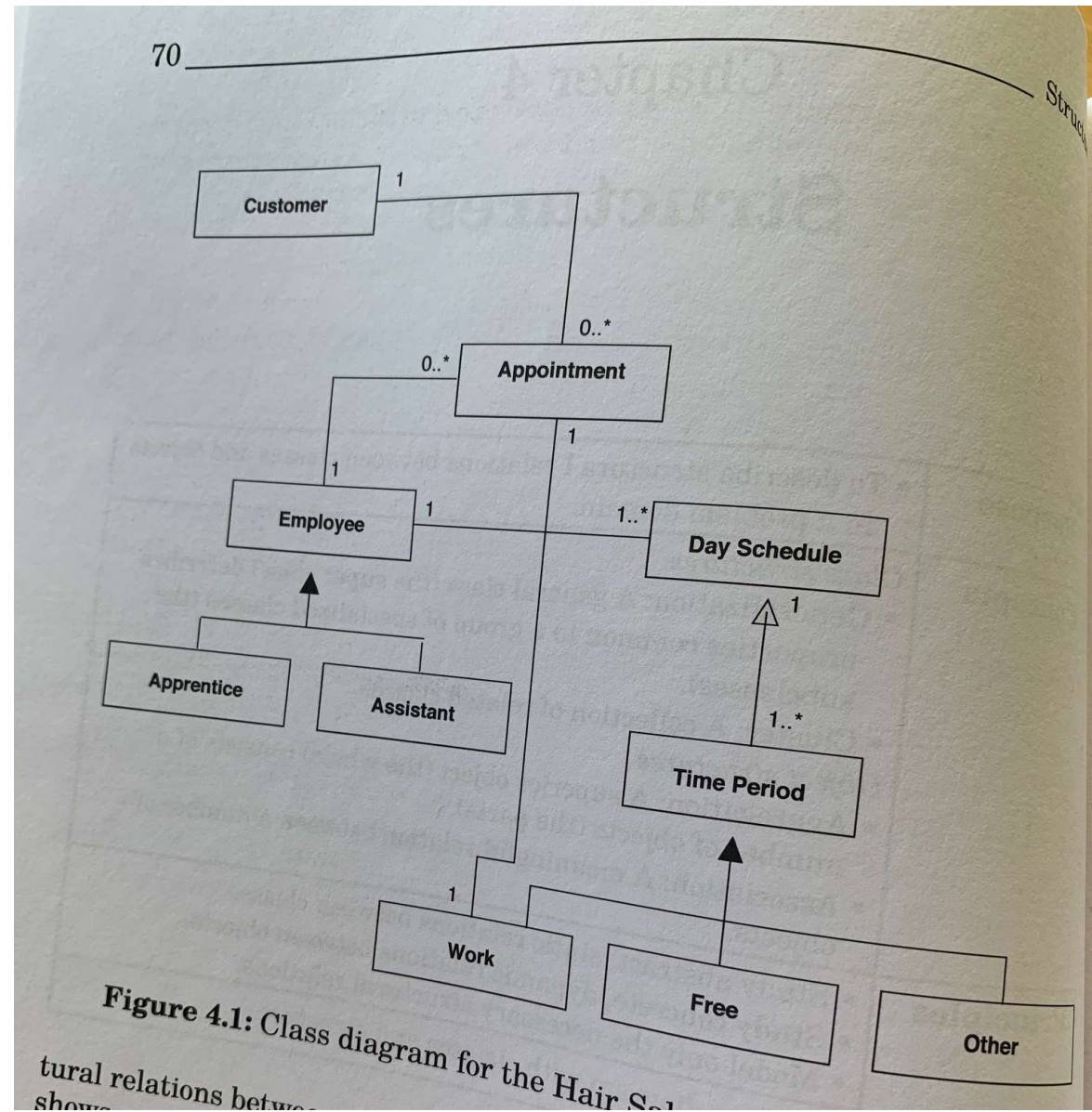


Different Models



A Part of a Model

Behind the scenes



Why Modelling?

- Provides Overview
- Supports Communication
- Prompts Questions
- Ensures Structure
- Supports Collaboration

UML

Unified Modelling Language, creating OO structures
since the late 90's

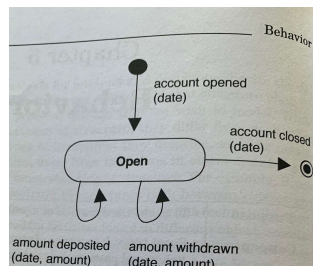
Standardized Representations

50

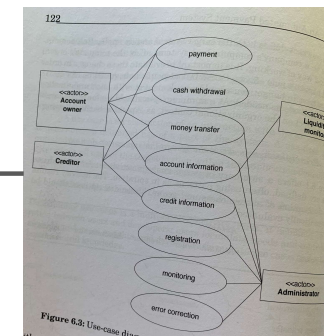
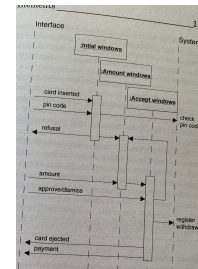
Events	Customer	Assisant	Apprentice	Appointment
reserved	✓			
cancelled	✓	✓		✓
treated	✓			
employed				✓
resigned		✓	✓	
graduated		✓	✓	
agreed		✓	✓	

Figure 3.1: Event table

Event tables



State charts



Use case diagrams

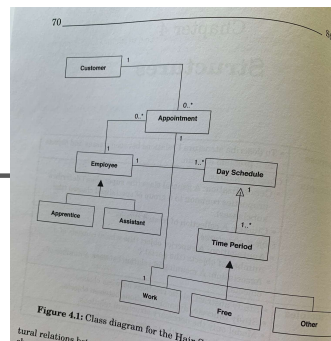
Sequence diagrams

Actors

Use cases	Account owner	Creditor	Administrator	Liquid assets
Payment	✓	✓		
Cash withdrawal	✓			
Money transfer	✓			
Account information	✓	✓	✓	
Credit information		✓	✓	
Registration			✓	
Monitoring			✓	
Error correction			✓	

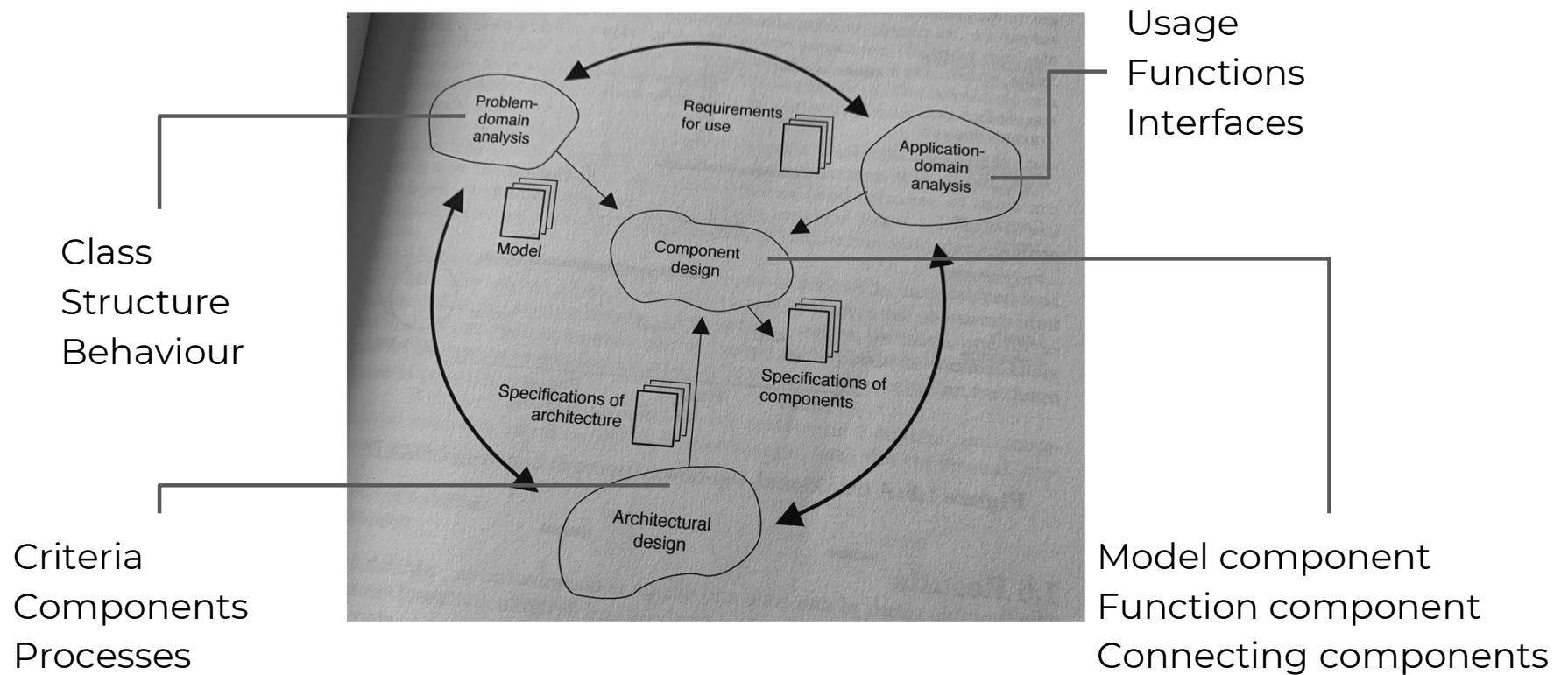
Actor tables

Actor tables

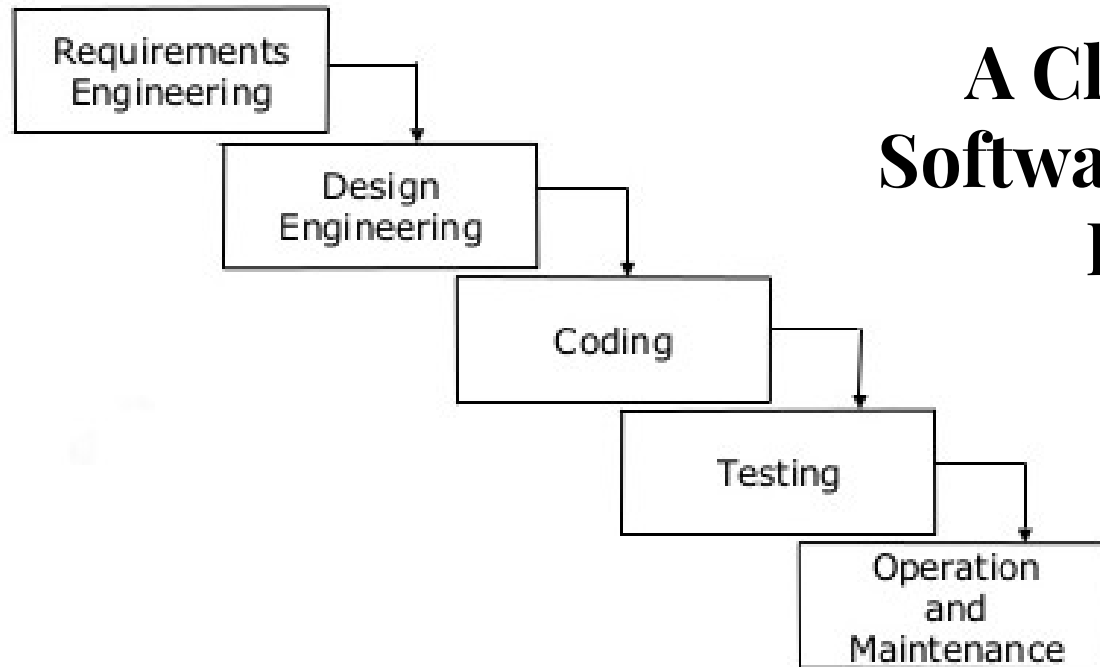


Class digrams

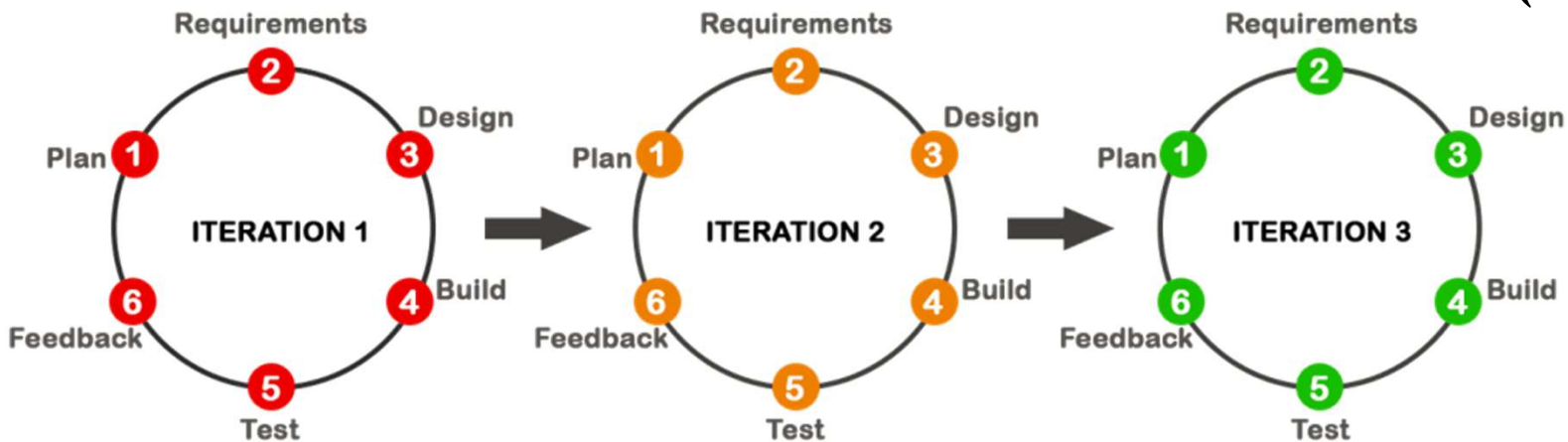
Main activities and results



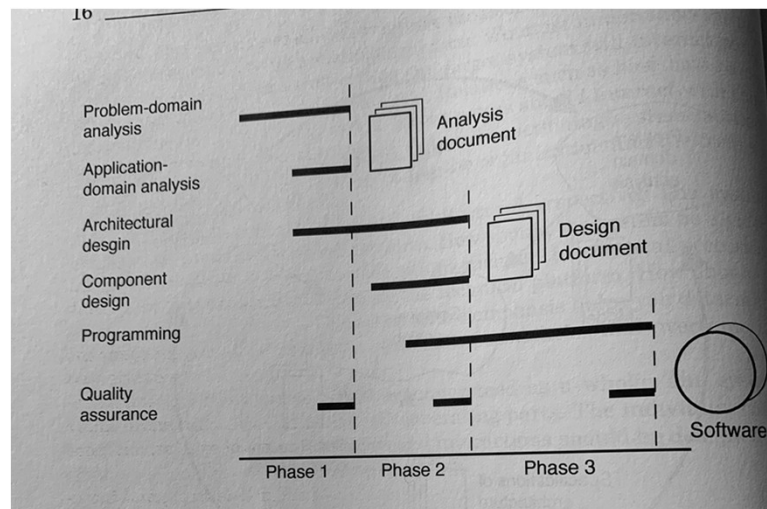
A Classic Sequential Software Development Lifecycle (SDLC)



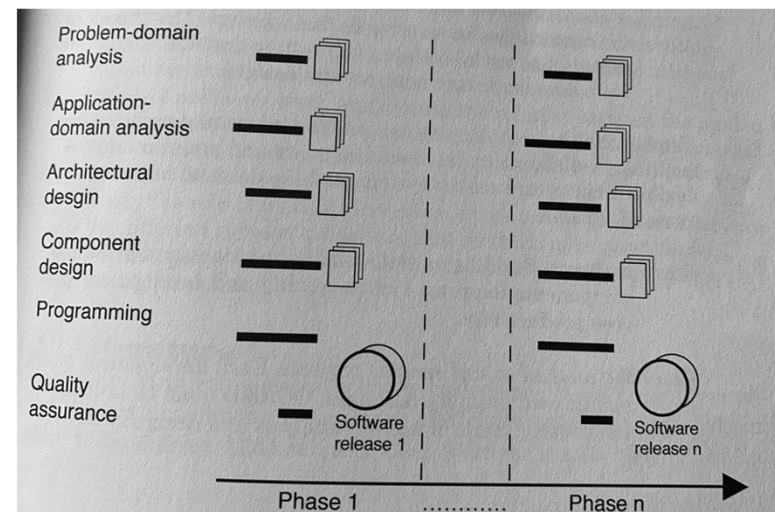
A Classic Agile Software Development Lifecycle (SDLC)



Processes as per the Course Book



Sequential

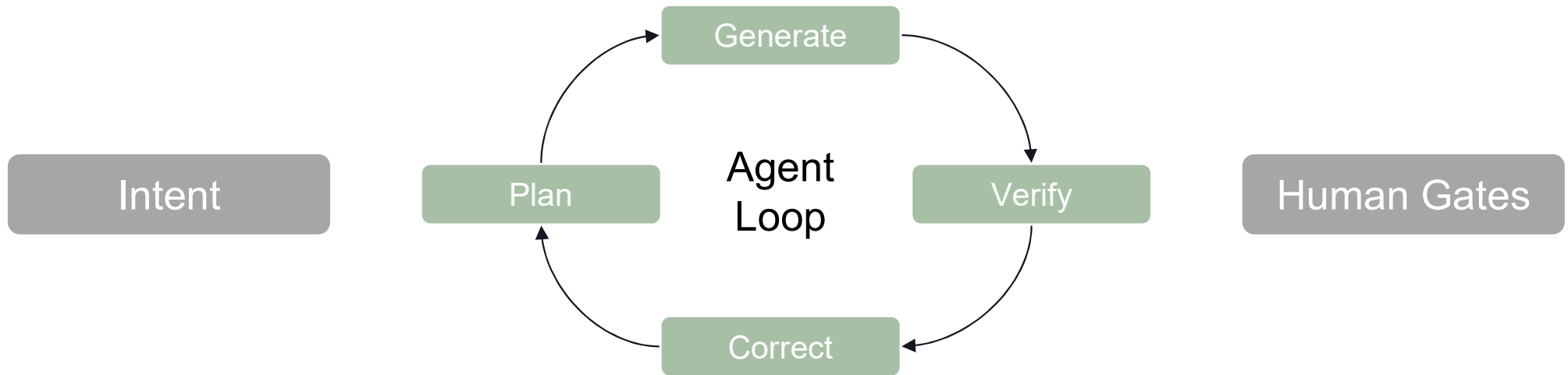


Iterative

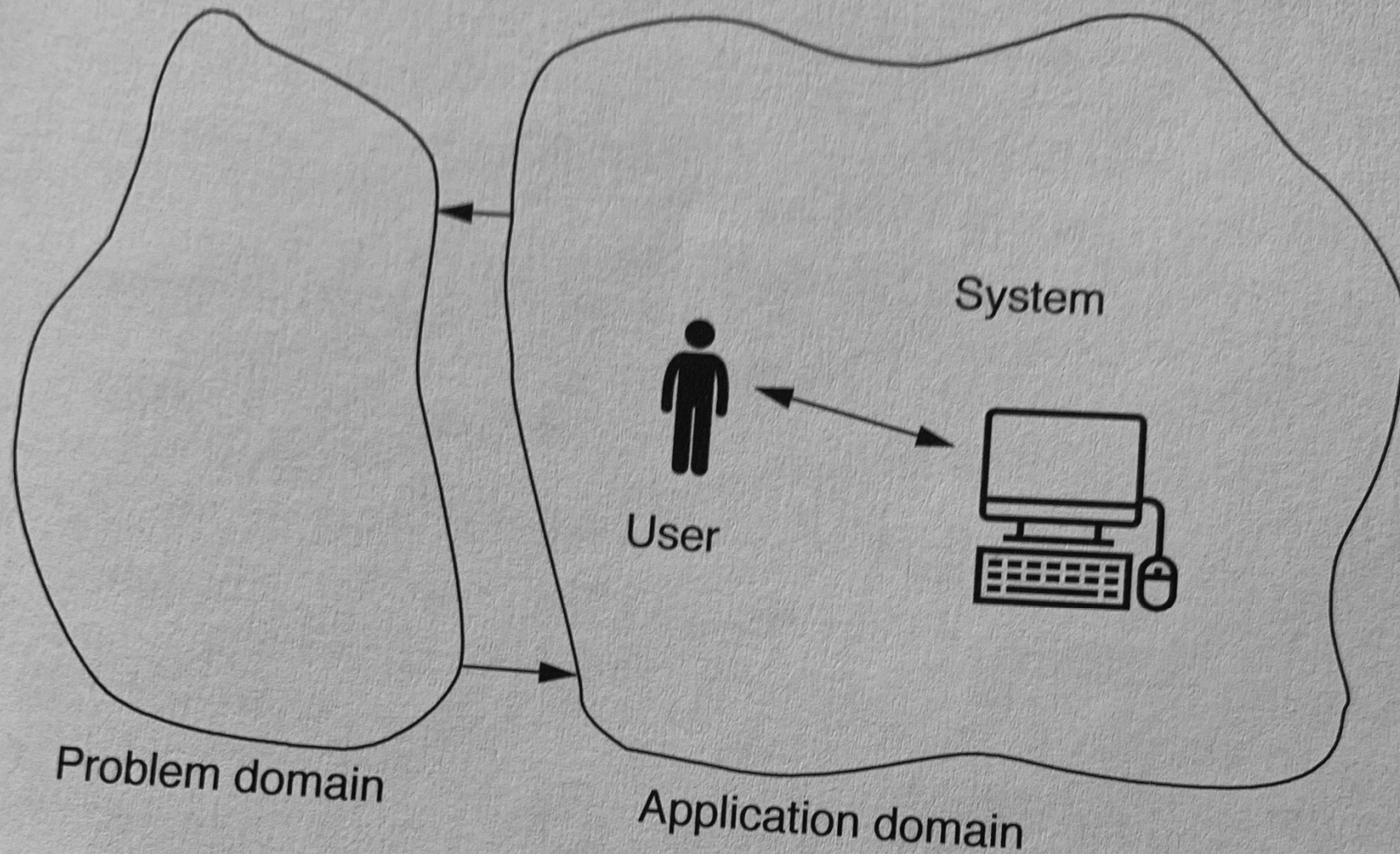
GAI

Claude Code, Open AI Codex, MS Co-Pilot etc.

AI Software Development Lifecycle (AI-DLC)

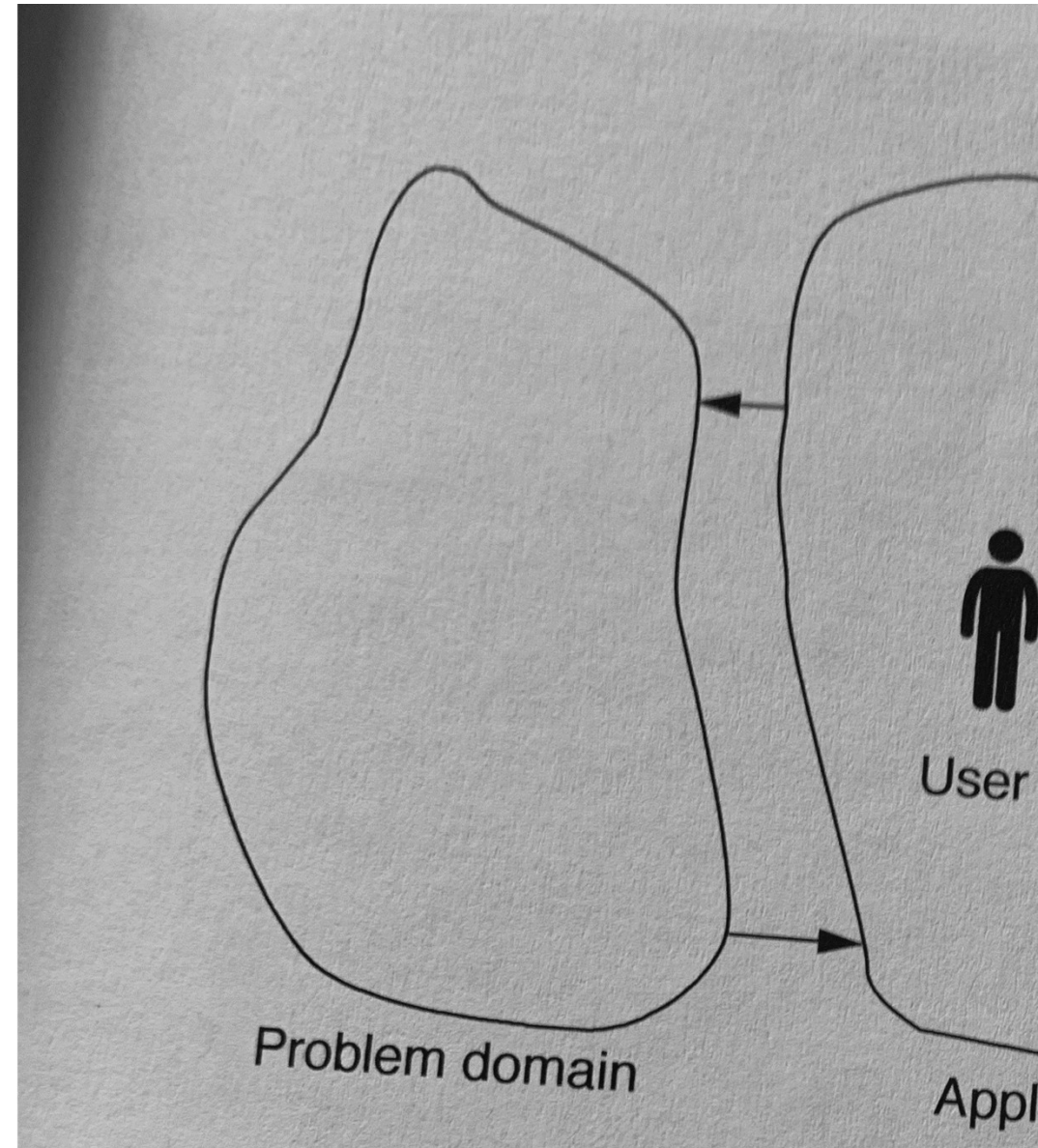


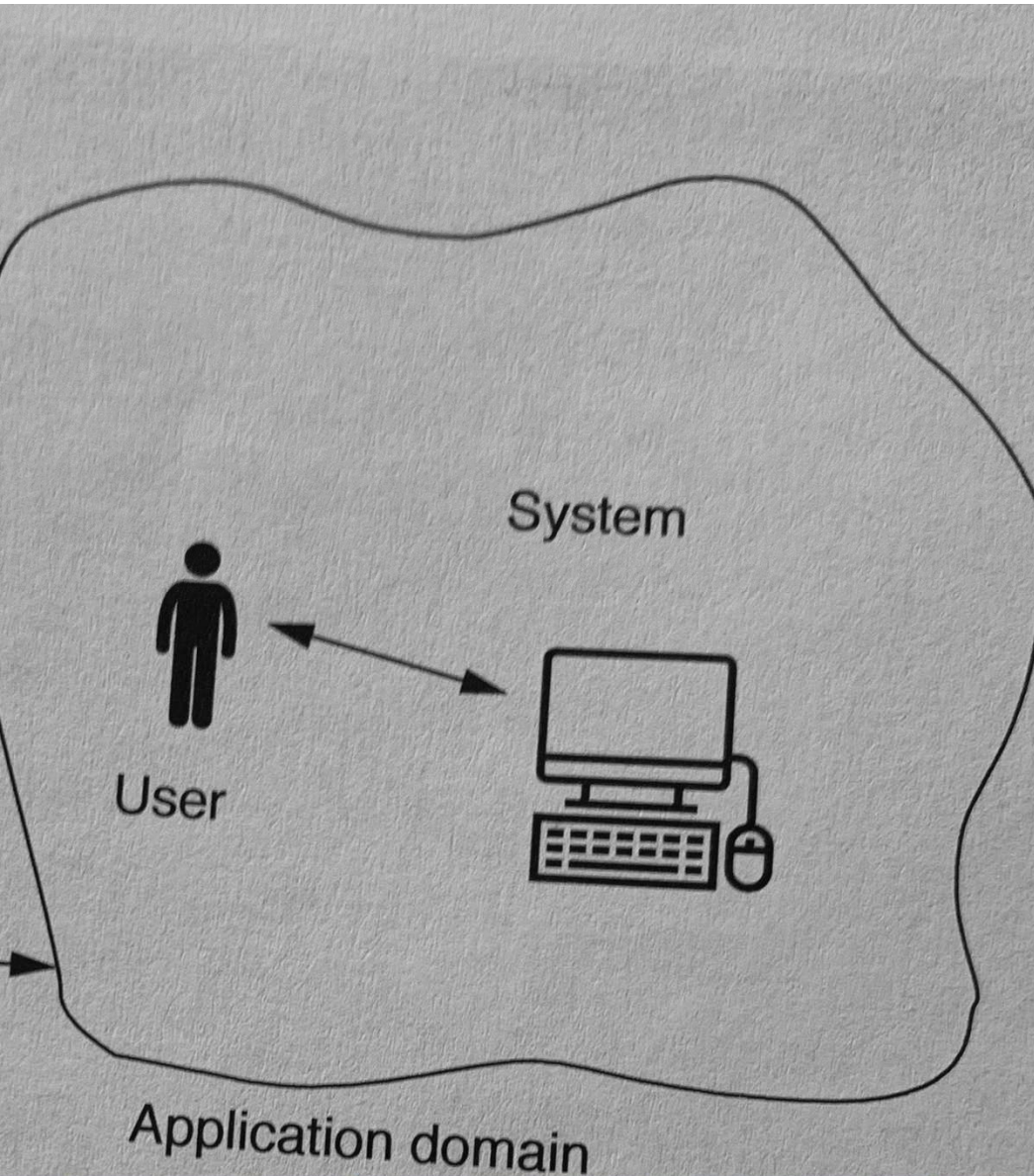
Problem domain and Application Domain



Problem Domain

Part of the context that is administered, monitored or controlled by a system





Application Domain

The organization that administrates, monitors or controls a problem domain



Problem Domain and Application Domain?

Why this distinction?

System Definition

A concise description of a computerized system expressed in natural language.

FACTOR criterion:

- **Functionality.** System functions to support the application domain tasks
- **Application domain.** Parts of an organization that administrate, monitor or control a problem domain
- **Conditions.** Conditions under which the system will be developed and used
- **Technology.** Technology used to develop the system and technology on which the system will operate
- **Objects.** Main objects in the problem domain (consider the types of objects, i.e. the classes)
- **Responsibility.** The systems overall responsibility in relation to its context

Objects and Classes

Consider: Objects that we can identify that should be administrated by the system



Movies

A class called movie is created to hold the information of all the movies that has been shown and will be shown in the movie theater. Each movie in the movie theater will be an instance / an object in the class.

EXAMPLE: A MOVIE THEATER



Objects and Classes (movie theater)

Classes A description of a collection of objects sharing structure, behavioral pattern, and attributes. Each class consists of a number of objects.	Objects An entity with identity, state, and behavior. An object belongs to a class.
Movie	Avatar, The Godfather, Hard to Kill
Customer	Sophie James, Olivia Rose, Michael David
Showing	Tuesday 20-11-2025 at 6pm Friday 10-10-2025 at 5pm Friday 10-10-2025 at 7pm
Ticket	VIP, Standard Adult, Standard Child
Theater	Nordic Film Aalborg, Nordic Film Aarhus

Problem Domain vs. Application Domain

Exercise 1.1

Imagine a system that is intended to administer teaching activities at individual departments of a university (Department of Computer Science, Electronic Systems etc.). It should be able to handle and organize all central administrative processes around students and lecturers as well as the planning and delivery of courses. The system will be used by semester secretaries, lecturers and students. Note however, that booking of rooms will be handled by a different centralised system as each department is able to book rooms across all other departments.

1. Identify objects relevant to consider for this system
2. Discuss which objects belong to the problem domain of this system, and what belongs to the application domain. Is there anything that belongs to both?

System Definition

Exercise 1.2

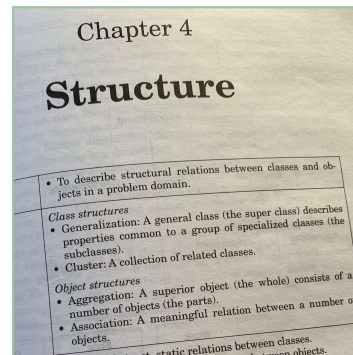
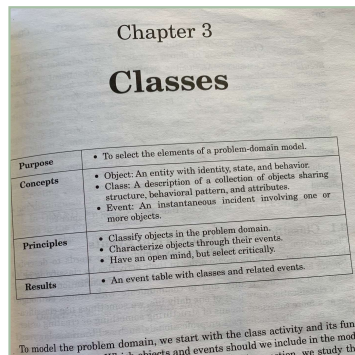
Make a system definition of the system to administer a university (same system as imagined in exercise 1.1). Use **FACTOR** to formulate the system definition.

System Definition – Semester Project

Exercise 1.3

Make a system definition of the system for your semester project. Use **FACTOR** to formulate the system definition.

Until we meet again



Supplementary material